

The Company within Social and Planetary Boundaries

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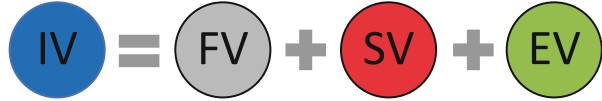
Overview

Society faces multiple sustainability challenges. On the environmental front, climate change, land-use change, biodiversity loss, freshwater shortages, and depletion of natural resources are destabilising the Earth system. On the social side, many people are afflicted by poverty, hunger, and lack of healthcare. *Sustainability* means that current and future generations have the resources needed, such as food, water, healthcare, and energy, without stressing the Earth system processes. The United Nations' Sustainable Development Goals (SDGs) are a guide for the transition towards a sustainable and inclusive economy.

Companies play an important role in that transition to a sustainable economy because social and environmental impacts are generated primarily in the corporate sector. Some companies will survive the transition by providing valuable solutions; others will not, as their competitive positions are eroded. Sustainability is therefore also about corporate survival. Responsible companies are increasingly adopting the goal of integrated value creation, which unites financial, social, and environmental value.

This raises the fundamental question in corporate finance: what is the objective of the company? The traditional objective is maximising profit, which boils down to maximising financial value for shareholders. This does not incentivise companies to act in a sustainable manner. An alternative view is to broaden the objective of the company to optimising integrated value (IV), which combines financial value (FV), social value (SV), and environmental value (EV). Figure 1.1 depicts the new paradigm of integrated value. In that way, the interests of current and future stakeholders are equal and aligned with sustainable development. Applying this new paradigm of integrated value is the real innovation of this corporate finance textbook.

This integrated approach to value has profound implications for corporate finance. It challenges conventional thinking and practices regarding various aspects of financial decision-making, including corporate investments, valuation, and capital structure.

Fig. 1.1 Integrated value

Learning Objectives

After you have studied this chapter, you should be able to:

- Analyse the planet's social and environmental challenges
- Identify and interpret the United Nations Sustainable Development Goals
- Define the transition towards a sustainable world
- Critically review the objective of the company
- Demonstrate the concept of integrated value
- Identify the challenges of incorporating sustainability into corporate finance

1.1 Social and Planetary Boundaries in a Full World

Our economic models were developed in an empty world, with an abundance of goods and services produced by nature (Daly & Farley, 2011). These models assumed that labour and capital were the scarce production factors to optimise in economic production, while nature and its services were freely available. Human society became dependent on fossil fuels and other non-renewable resources. Technological advances allowed the unprecedented production of consumer goods, spurring economic and population growth. Urbanisation led to a reduction in arable land, driving further deforestation.

More recently, we have started to realise that natural resources are finite. In the early 1970s, the Club of Rome warned that the Earth system cannot support our rates of economic and population growth much beyond 2100. Its report *Limits to Growth* argues that humankind can create a society in which it lives indefinitely on earth. This requires a balance between population and production, for which humankind needs to impose limits on its production of material goods (Meadows et al., 1972).

Another step in awareness was the United Nations' (UN) Brundtland Report (1987), which argues that '...the *environment* is where we live; and *development* is what we all do in attempting to improve our lot within that environment. The two are inseparable'. The report defines *sustainable development* as '*development that meets the needs of the present without compromising the ability of future generations to meet their own needs*'. The Brundtland report stresses that sustainability is about the future.

Climate change is one of the largest environmental risks affecting society. The United Nations Framework Convention on Climate Change (UNFCCC) is an international environmental treaty. Since 1995, it has been organising Conferences of the Parties (COP) to assess progress in dealing with climate change. In the 2015 Paris Agreement on climate change (COP21), countries reconfirmed the target of limiting the rise in global average temperatures relative to those in the preindustrial world to 2 °C (two degrees Celsius) and to pursue efforts to limit the temperature increase to 1.5 °C (UNFCCC, 2015). Achieving this would ensure that the stock of carbon

dioxide (CO₂) and other greenhouse gases in the atmosphere does not exceed a certain limit. The Intergovernmental Panel on Climate Change (2023) estimates that the remaining carbon budget amounts to 1150 gigatonnes of CO₂ from 2020 onwards (for a probability of 67% for limiting global warming to 2 °C). If current global carbon emissions (approximately 40 gigatonnes a year) are not drastically cut, the 2 °C limit would be reached around the year 2048.¹

The most pressing environmental and social challenges include climate risk, land-system change, biodiversity loss, green water, nitrogen and phosphorus flows, poverty, food, and health problems. Our economic system creates these environmental and social impacts on society; they are inseparable from production decisions. To highlight the tension between unbridled economic growth and sustainable development, we provide two examples. Box 1.1 describes the Deepwater Horizon oil spill. Box 1.2 illustrates the impact of the collapse of a factory building in Bangladesh. Both examples involve an underinvestment in safety to increase short-term profits.

Box 1.1: The Deepwater Horizon Oil Spill

Oil began to spill from the Deepwater Horizon drilling platform on 20 April 2010, in the BP-operated Macondo Prospect in the Gulf of Mexico. An explosion on the drilling rig led to the largest accidental marine oil spill in the history of the petroleum industry. The US Government estimated the total discharge at 4.9 million barrels.

A massive response ensued to protect beaches, wetlands, and estuaries from the spreading oil. Oil clean-up crews worked on 55 miles of the Louisiana shoreline until 2013. The months-long spill caused extensive damage to marine and wildlife habitats and the fishing and tourism industries.

Investigation pointed to defective cement on the well, laying the fault mostly with BP, but also with rig operator Transocean and contractor Halliburton. In 2011, a National Commission (2011) likewise blamed BP and its partners for a series of cost-cutting decisions and an inadequate safety system; it also concluded that the spill resulted from ‘*systemic*’ root causes and that without ‘*significant reform in both industry practices and government policies, might well recur*’.

¹The carbon budget for the more stringent 1.5 °C limit is 500 gigatonnes of CO₂ from 2020 onwards. That would be reached around the year 2032.

Box 1.2: Rana Plaza Factory Collapse

The Rana Plaza collapse was a disastrous structural failure of an eight-storey commercial building on 24 April 2013 in Bangladesh. The collapse of the building caused 1129 deaths, while approximately 2500 injured people were rescued alive from the building. It is considered the deadliest garment factory accident in history and the deadliest accidental structural failure in modern human history.

The building contained clothing factories, a bank, apartments, and several shops. The shops and the bank on the lower floors were immediately closed after cracks were discovered in the building. The building's owners ignored warnings to evacuate the building after cracks in the structure appeared the day before the collapse. Garment workers, earning €38 a month, were ordered to return the following day, and the building collapsed during the morning rush-hour.

The factories manufactured clothing for brands including Benetton, Bonmarché, the Children's Place, El Corte Inglés, Joe Fresh, Monsoon Accessorize, Mango, Matalan, Primark, and Walmart.

1.1.1 Planetary Boundaries

There can be no Plan B, because there is no Planet B.

Former United Nations Secretary-General, Ban Ki-moon

There is increasing evidence that human activities are affecting the Earth system, threatening the planet's future liveability. The planetary boundaries framework of Steffen et al. (2015) defines a safe operating space for humanity within the boundaries of nine productive ecological capacities of the planet. The framework is based on the intrinsic biophysical processes that regulate the stability of the Earth system on a planetary scale. The green zone in Fig. 1.2 is the safe operating space; orange represents the zone of uncertainty (increasing risk) and red (dark orange) indicates the zone of high risk. Table 1.1 specifies the control variables and quantifies the ecological ceilings.

Applying the *precautionary principle*, the *planetary boundary* itself lies at the intersection of the green and orange zones. To illustrate how the framework works, we look at the control variable for climate change, the atmospheric concentration of greenhouse gases. The zone of uncertainty ranges from 350 to 450 parts per million (ppm) of carbon dioxide. We crossed the planetary boundary of 350 ppm in 1988, with a level of 420 ppm in early 2023 (see Table 1.1). The upper limit of 450 ppm is consistent with the goal (at a fair chance of 67%) to limit global warming to 2 °C above the preindustrial level and lies at the intersection of the orange and red zones.

Another example in the orange zone of increasing risk is land-system change. The control variable is the area of forested land as a proportion of forest-covered land

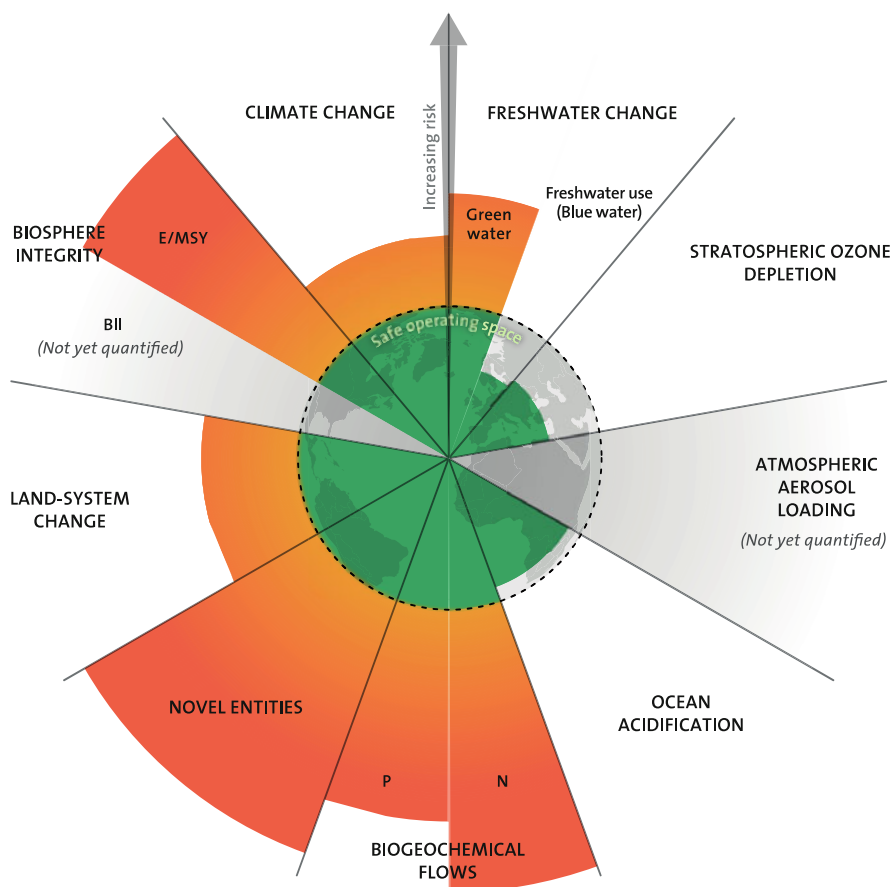


Fig. 1.2 The planetary boundaries. Source: Azote for Stockholm Resilience Centre, based on analysis in Persson et al. (2022), Steffen et al. (2015), and Wang-Erlandsson et al. (2022)

prior to human alteration. The planetary boundary is at 75% (safe minimum), while we are currently at 62%, and the percentage is falling (worsening).

The current *linear production and consumption system* is based on the extraction of raw materials (take), processing them into products (make), consumption (use), and disposal (waste). Traditional business models centred on a linear system assume the on-going availability of unlimited and cheap natural resources. This is increasingly risky because non-renewable resources, such as fossil fuels, minerals, and metals, are increasingly under pressure, while potentially renewable resources, such as forests, rivers, and prairies, are declining in their extent and regenerative capacity.

Moreover, the use of fossil fuels in the linear production and consumption system overburdens the Earth system as a natural sink (absorbing pollution). Baseline scenarios (i.e., those without mitigation) for climate change result in global warming in 2100 at approximately 3.0 °C compared to the preindustrial level (Intergovernmental Panel on Climate Change, 2023). Furthermore, food production leads to

Table 1.1 The ecological ceiling and its indicators of overshoot

Earth system pressure	Control variable	Planetary boundary	Current value and trend
Climate change	Atmospheric carbon dioxide concentration; parts per million (ppm)	At most 350 ppm	420 ppm and rising (worsening)
Biosphere loss	Genetic diversity: rate of species extinction per million species per year	At most 10	Around 100–1000 and rising (worsening)
	Functional diversity: biodiversity intactness index (BII)	Maintain BII at 90%	84% applied to southern Africa only
Land-system change	Area of forested land as a proportion of forest-covered land prior to human alteration	At least 75%	62% and falling (worsening)
Freshwater change	Green water: percentage of ice-free land area on which root-zone soil moisture is too low (or too high)	—	—
	Blue water consumption; cubic kilometres per year	At most 4000 km ³	Around 2600 km ³ and rising (intensifying)
Biochemical flows	Phosphorus applied to land as fertiliser; millions of tons per year	At most 6.2 million tons	Around 14 million tons and rising (worsening)
	Reactive nitrogen applied to land as fertiliser; millions of tons per year	At most 62 million tons	Around 150 million tons and rising (worsening)
Ocean acidification	Average saturation of aragonite (calcium carbonate) at the ocean surface, as a percentage of preindustrial levels	At least 80%	Around 84% and falling (intensifying)
Air pollution	Aerosol optical depth (AOD); much regional variation, no global level yet defined	—	—
Ozone layer depletion	Concentration of ozone in the stratosphere; in Dobson Units (DU)	At least 275 DU	283 DU and rising (improving)
Novel entities	Production volume of plastics	—	—
	Production volume of hazardous chemicals	—	—

Source: Updated from Persson et al. (2022), Steffen et al. (2015), and Wang-Erlandsson et al. (2022)

biodiversity loss because of the conversion of natural forests to agriculture (land-system change).

With this linear economic system, we are crossing planetary boundaries beyond which human activities might destabilise the Earth system. In particular, the planetary boundaries of climate change, land-system change (deforestation and land erosion), biodiversity loss (terrestrial and marine), green water (needed for vegetation), novel entities (plastics and chemicals), and biochemical flows (nitrogen and phosphorus, mainly because of intensive agricultural practices) have been crossed

(see Fig. 1.2). A timely transition towards an economy based on sustainable production and consumption, including the use of renewable energy, reuse of materials and land restoration, can mitigate these risks to the stability of the Earth system.

1.1.2 Social Foundations

Mass production in a competitive economic system has led to long working hours, underpayment, and child labour, first in the developed world and later relocated to the developing world. Human rights provide the essential social foundation for all people to lead lives of dignity and opportunity. Human rights norms assert the fundamental moral claim each person has to life's essential needs, such as food, water, healthcare, education, freedom of expression, political participation, and personal security. Raworth (2017) defines the *social foundations* as the twelve top social priorities, grouped into three clusters, focused on enabling people to be:

1. Well: through food security, adequate income, improved water and sanitation, housing and healthcare
2. Productive: through education, decent work, and modern energy services
3. Empowered: through networks, gender equality, social equity, political voice, and peace and justice

While these social foundations only set out the minimum of every human's claims, sustainable development envisions people and communities prospering beyond this, leading lives of creativity and thriving. Sustainable development combines the concept of planetary boundaries with the complementary concept of social foundations or boundaries (Sachs, 2015). *Sustainability* means that current and future generations have the resources needed, such as food, water, healthcare and energy, without stressing processes within the Earth system.

Many people are still living below the foundational social boundaries of no hunger, no poverty (a minimum income of \$2.15 a day), access to education and access to clean cooking facilities (see Table 1.2). Political participation, which is the right of people to be involved in decisions that affect them, is a basic value of society. The UN's Universal Declaration of Human Rights states that '*recognition of the inherent dignity and of the equal and inalienable rights of all members of the human family is the foundation of freedom, justice and peace in the world*'. Decent work can lift communities out of poverty and underpins human security and social peace. The 2030 Agenda for Sustainable Development (see Sect. 1.2 below) places decent work for all people at the heart of policies for sustainable and inclusive growth and development. Decent work has several dimensions: a basic living wage (which depends on a country's basket of basic goods), no discrimination (e.g. gender, race, or religion), no child labour, health and safety, and freedom of association.

From a societal perspective, it is important for business to respect these social foundations and to ban underpayment, child labour, and human rights violations.

Table 1.2 The social foundation and its indicators of shortfall

Dimension	Illustrative indicator (per cent of global population unless otherwise stated)	%	Year
Food	Population undernourished	11	2014–16
Health	Population living in countries with under-five mortality rate exceeding 25 per 1000 live births	46	2015
	Population living in countries with life expectancy at birth of less than 70 years	39	2013
Education	Adult population (aged 15+) who are illiterate	15	2013
	Children aged 12–15 out of school	17	2013
Income and work	Population living on less than the international poverty limit of \$2.15 a day	29	2012
	Proportion of young people (aged 15–24) seeking but not able to find work	13	2014
Water and sanitation	Population without access to improved drinking water	9	2015
	Population without access to improved sanitation	32	2015
Energy	Population lacking access to electricity	17	2013
	Population lacking access to clean cooking facilities	38	2013
Networks	Population stating that they are without someone to count on for help in times of trouble	24	2015
	Population without access to the Internet	57	2015
Housing	Global urban population living in slum housing in developing countries	24	2012
Gender equality	Representation gap between women and men in national parliaments	56	2014
	Worldwide earnings gap between women and men	23	2009
Social equity	Population living in countries with a Palma ratio of 2 or more (the ratio of the income share of the top 10% of people to that of the bottom 40%)	39	1995–2012
Political voice	Population living in countries scoring 0.5 or less out of the 1.0 in the Voice and Accountability Index	52	2013
Peace and justice	Population living in countries scoring 50 or less out of 100 in the Corruption Perceptions Index	85	2014
	Population living in countries with a homicide rate of 10 or more per 10,000	13	2008–13

Source: Updated from Raworth (2017) and World Bank

Social regulations have been introduced in developed countries, but violations still occur in developing countries. A case in point is the use of child labour in factories in developing countries producing consumer goods, such as clothes and shoes, to be sold by multinational companies in developed countries. These factories often lack basic worker safety features (Box 1.2). Another example is the violation of the human rights of indigenous peoples, often in combination with land degradation and pollution, by extractive companies in the exploration and exploitation of fossil fuels, minerals, and other raw materials.

1.2 Sustainable Development Goals

To guide the transition towards a sustainable and inclusive economy, the United Nations has developed the 2030 Agenda for Sustainable Development (UN, 2015). This includes the 17 UN Sustainable Development Goals (SDGs), which aim to stimulate action over the period of 2015–2030 in areas of critical importance for humanity and the planet. Box 1.3 explains the SDGs.

Box 1.3: The Sustainable Development Goals

The following are the 17 UN Sustainable Development Goals (UN, 2015):

Economic Goals

- Goal 8. Promote sustained, inclusive, and sustainable economic growth, full and productive employment and decent work for all
- Goal 9. Build resilient infrastructure, promote inclusive and sustainable industrialisation, and foster innovation
- Goal 10. Reduce inequality within and among countries
- Goal 12. Ensure sustainable consumption and production patterns

Societal Goals

- Goal 1. End poverty in all its forms everywhere
- Goal 2. End hunger, achieve food security and improved nutrition, and promote sustainable agriculture
- Goal 3. Ensure healthy lives and promote well-being for all at all ages
- Goal 4. Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all
- Goal 5. Achieve gender equality and empower all women and girls
- Goal 7. Ensure access to affordable, reliable, sustainable, and modern energy for all
- Goal 11. Make cities and human settlements inclusive, safe, resilient, and sustainable
- Goal 16. Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable, and inclusive institutions at all levels

Environmental Goals

- Goal 6. Ensure availability and sustainable management of water and sanitation for all
- Goal 13. Take urgent action to combat climate change and its impacts

(continued)

Box 1.3 (continued)

- Goal 14. Conserve and sustainably use the oceans, seas, and marine resources for sustainable development
- Goal 15. Protect, restore, and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, halt and reverse land degradation and halt biodiversity loss

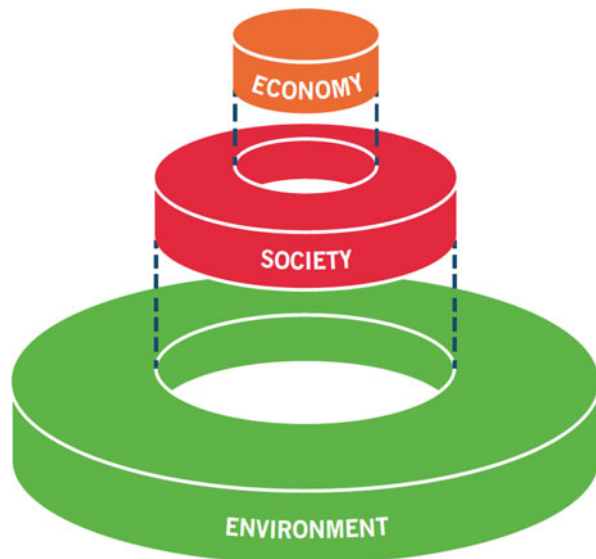
Overall Goal

- Goal 17. Strengthen the means of implementation and revitalise the Global Partnership for Sustainable Development

To facilitate implementation, the 17 high-level goals are specified in 169 targets (see <https://sustainabledevelopment.un.org/topics/sustainabledevelopmentgoals>).

Following Rockström and Sukhdev (2016), we classify the SDGs according to several levels: the economy, society, and environment (Fig. 1.3). Nevertheless, we stress that the SDGs are interrelated. A case in point is the move to sustainable consumption and production (Economic Goal 12) and sustainable cities (Societal Goal 11), which are instrumental to combat climate change (Environmental Goal 13). Another example is appropriate income and decent work for all (Economic Goal 8), which is instrumental in attaining Societal Goals 1–4. Through a living wage, households can afford food, healthcare, and education for their family.

Fig. 1.3 Sustainable development challenges at different levels. Source: Adapted from Rockström and Sukhdev (2016)



1.2.1 Global Strategy

The UN SDGs are the global strategy of governments under the auspices of the United Nations and provide direction towards (future) government policies, such as the regulation and taxation of environmental and social challenges. This strategy is boosted by technological change (e.g. the development of solar and wind energy and electric cars at decreasing cost or the development of drip irrigation systems), which supplements government policies (e.g. carbon pricing or water pricing). Some companies are preparing for this transition (future makers) and are part of the solution (Mercer, 2015). Other companies are waiting for the transition to unfold before acting (future takers). A final category of companies is unaware of this transition and continues business as usual.

We, as authors, attach a positive probability to the scenario that the SDGs are largely met. Our observation is based on the success of the earlier Millennium Development Goals in reducing poverty, hunger, and child death rates in Southeast Asia and Latin America but less so in Africa. Of course, opinions can, and do, differ about the probability that the transition towards a sustainable economy will largely succeed. However, the ‘business-as-usual’ scenario, which assumes no transition, is highly implausible. While the pathway and the speed of the transition are uncertain and may even be erratic, with failures along the way, the sustainable development agenda gives direction to thinking about the future. This book is about the role corporate finance can play in shaping this future and making production and consumption more sustainable via future-proof investment decisions.

The UN SDGs address challenges at the level of the economy, society, and the environment (or biosphere). Figure 1.3 illustrates the three levels and the ranking between them. A liveable planet is a precondition or foundation for humankind to thrive. A cohesive and inclusive society is needed to organise production and consumption to ensure enduring prosperity for all. In their seminal book *Why nations fail*, Acemoglu and Robinson (2012) show that political institutions that promote inclusiveness generate prosperity. Inclusiveness allows everyone to participate in economic opportunities. Reducing social inequalities (SDG 10) is thus an important goal. Of course, there can be resource conflicts: unequal communities may disagree over how to share and finance public goods. These conflicts, in turn, break social ties and undermine the formation of trust and social cohesion (Barone & Mocetti, 2016; Berger, 2018).

1.2.2 System Perspective

While it is tempting to start working on partial solutions at each level, the environmental, societal, and economic challenges are interlinked. It is important to embrace an integrated social-ecological system perspective (Norström et al., 2014). Such an integrated system perspective highlights the dynamics that such systems entail, including the role of ecosystems in sustaining human well-being, cross-system interactions, and uncertain thresholds.

Holling (2001) describes the process of sustainable development as embedded cycles with adaptive capacity. A key element of adaptive capacity is the *resilience* of the system to deal with unpredictable shocks (which is the opposite of the vulnerability of the system). Complex systems feature adaptive cycles that aggregate resources and periodically restructure to create opportunities for innovation. However, some systems are maladaptive and trigger, for example, a poverty trap or land degradation (i.e., the undermining of the quality of soil as a result of human behaviour or severe weather conditions). Holling (2001) concludes that ecosystem management via incremental increases in efficiency does not work. For transformation, ecosystem system management must build and maintain ecological resilience as well as social flexibility to cope, innovate, and adapt.

1.2.2.1 Examples of Cross-system Interactions and Uncertain Thresholds

A well-known example of cross-system interaction is the linear production of consumption goods at the lowest cost contributing to ‘economic growth’ while depleting natural resources, using child labour and producing carbon emissions and other waste. In this book, we use carbon emissions as a shorthand for all greenhouse gas emissions, which include carbon dioxide CO₂, methane CH₄, and nitrous oxide N₂O.

Another example of cross-system interaction is climate change leading to increasingly intense weather-related disasters, such as storms, flooding, and droughts. The low- and middle-income countries around the equator are especially vulnerable to these extreme weather events, which could damage a large part of their production capacity. The temporary loss of tax revenues and increase in expenditure to reconstruct factories and infrastructure might put vulnerable countries into a downwards fiscal and macroeconomic spiral with an analogous increase in poverty. Social and ecological issues are thus interconnected, whereby the poor in society are more dependent on ecological services and are less protected against ecological hazards.

An example of an uncertain threshold combined with feedback dynamics is the melting of the Greenland ice sheet. New research has found that it is more vulnerable to global warming than previously thought. Robinson et al. (2012) calculate that a 0.9 °C increase in global temperature from today’s levels could lead the Greenland ice sheet to melt completely. Such melting would create further climate feedback in the Earth’s ecosystem because melting the polar icecaps could increase the pace of global warming (by reducing the refraction of solar radiation, which is 80% from ice, compared with 30% from bare earth and 7% from the sea) as well as rising sea levels. These feedback mechanisms are examples of tipping points and shocks, which might happen.

Another example of cross-system interaction between several planetary boundaries is biodiversity loss. Box 1.4 shows the direct drivers of biodiversity loss.

Box 1.4: Direct Drivers of Biodiversity Loss

There are five direct drivers of biodiversity loss:

1. **Climate change**, where a change in climate destabilises ecosystems
2. **Invasive species**, where animals or plants have been moved to places where they damage existing ecosystems, e.g., Japanese knotweed
3. **Land-use change**, such as cutting down a forest to make way for agriculture
4. **Overexploitation of natural resources**, where a resource is used up faster than it can be replaced, e.g., overfishing
5. **Pollution of air, land, or water**, such as overuse of fertiliser containing phosphorus and nitrogen

Source: CISL (2021).

We cannot understand the sustainability of organisations in isolation from the socioecological system in which they are embedded: what are the thresholds, sustainability priorities, and feedback loops? Moreover, we should consider not only the socioecological impact of individual organisations but also the aggregate impact of organisations at the system level. The latter is relevant for sustainable development.

1.3 The Objective of the Company

To discuss the role of companies in sustainability, we first need to establish the objective of the company. The classical shareholder model in corporate finance argues that companies should maximise *shareholder value* (Jensen, 2002). In contrast, the stakeholder model argues that large companies should act in the interests of financial as well as social stakeholders and optimise *stakeholder value* (Magill et al., 2015). The integrated model states that companies should optimise *integrated value*, which combines financial, social, and environmental value (Schoenmaker & Schramade, 2019). The choice of the value maximisation function has consequences for decision-making on corporate investment.

To classify the different corporate finance models, Fig. 1.4 shows our framework for managing sustainable development. At the level of the economy, the financial return and risk trade-off is optimised. This financial orientation supports the idea of profit maximisation by companies. Next, at the societal level, the impact of investment and business decisions on society is optimised. Finally, at the level of the environment, the environmental impact is optimised. As we have argued in Sect. 1.2, there are interactions between the levels. It is thus important to choose an appropriate combination of the financial, social, and environmental aspects.

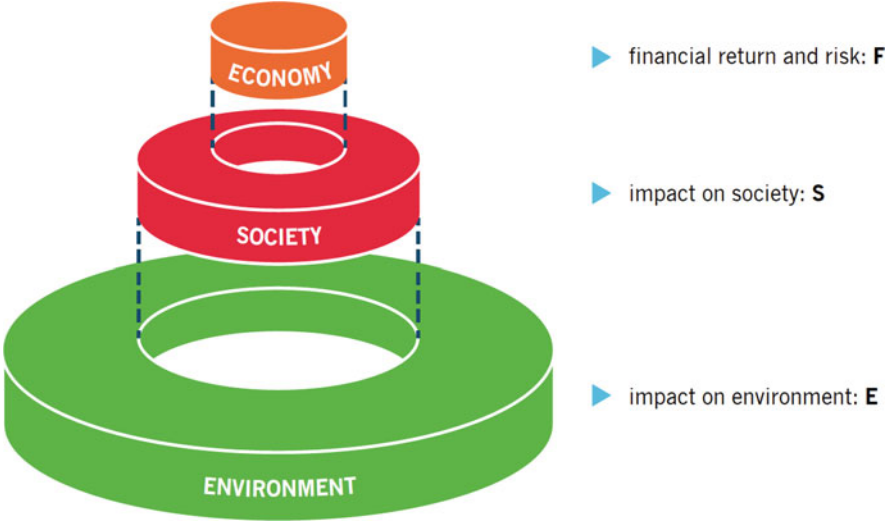


Fig. 1.4 Managing sustainable development. Source: Adapted from Schoenmaker and Schramade (2019)

Table 1.3 Corporate finance models

Corporate finance models	Value created	Main stakeholders	Ranking of factors	Optimisation of value V
Shareholder model	Shareholder value	Shareholders	FV	$\text{Max } V = \text{FV}$
Refined shareholder model	Shareholder value	Shareholders	FV SV & EV to extent they affect FV	$\text{Max } V = \text{FV} + b \cdot \text{SV} + c \cdot \text{EV}$ $0 < b, c \ll 1$
Stakeholder model	Stakeholder value	Current stakeholders	$\text{STV} = \text{FV} + \text{SV}$	$\text{Max } V = \text{FV} + b \cdot \text{SV}$ $b = 1$
Integrated model	Integrated value	Current and future stakeholders	$\text{IV} = \text{FV} + \text{SV} + \text{EV}$	$\text{Max } V = \text{FV} + b \cdot \text{SV} + c \cdot \text{EV}$ $b, c = 1$

Note: V, value; FV, financial value; SV, social value; EV, environmental value; STV, stakeholder value; IV, integrated value. In the refined shareholder model, SV and EV are relevant to the extent to which they affect FV
Source: Adapted from Schoenmaker and Schramade (2019)

Corporate finance has not yet caught up with the broader notion of business sustainability over the last decades. Table 1.3 shows the corporate finance models across four aspects:

- (i) The value created
- (ii) The main stakeholders

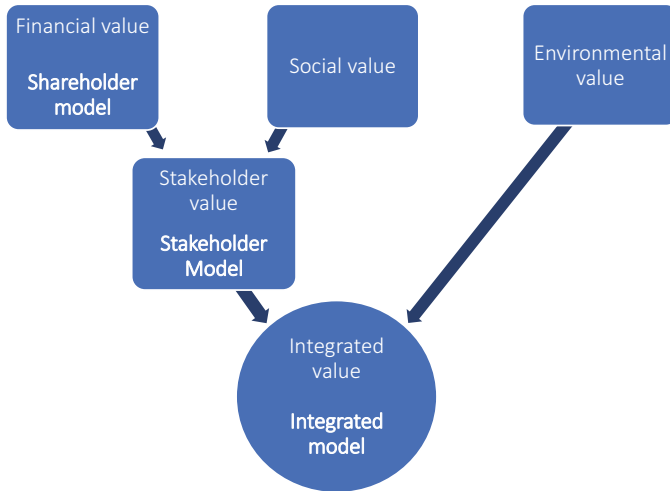


Fig. 1.5 Stages of corporate finance

- (iii) The ranking of the three factors and
- (iv) The optimisation method

Figure 1.5 highlights the broadening of corporate finance from *shareholder value* to *stakeholder value* and finally *integrated value*. The four stages of corporate finance in Table 1.3 are discussed one after another below. Figure 1.5 shows that the stages move from financial value to financial and social values combined and finally to financial, social, and environmental values equally (the ranking of factors in the fourth column of Table 1.3).

1.3.1 The Shareholder Model

In traditional corporate finance, the goal of the company is to maximise its financial value (Berk & DeMarzo, 2017; Brealey et al., 2020). This is the value of the securities (i.e. stocks and bonds) provided by the financiers (i.e. shareholders and creditors). Shareholders are deemed most important because they are residual, noncontractual claimants. They are paid after all contractual claims to other stakeholders, such as creditors, employees, customers, and government, are paid. Shareholders maximise financial value FV after the other stakeholders are satisfied. The formula for company value FV_0 at $t = 0$ is as follows:

$$FV_0 = \frac{CF_1}{(1+r)} + \frac{CF_2}{(1+r)^2} + \dots + \frac{CF_N}{(1+r)^N} \quad (1.1)$$

where r is the discount rate used to determine the present value at $t = 0$ of a future cash flow CF_t . Future cash flows are discounted from $t = 1$ to N . The cash flows are

those left to be distributed to financiers after all investments have been made. It is calculated as cash from operations minus cash into investments.

Corporate finance theory separates finance and ethics. Traditional finance is consistent with the argument of Friedman (1970) that '*the business of business is business*'. In this view, it is the task of the government to address social and environmental concerns. It is second nature for corporate managers to think and communicate in NPV terms (based on financial cash flows). This naturally affects the functions of valuation, allocation, and performance measurement in the investment process. It also affects how sustainability is integrated, how valuations are conducted, what investment approaches are favoured, and the role of corporate managers.

Where social and environmental *impacts* are important, a narrow focus on shareholder value can create scope for managers making morally dubious decisions. For example, maximising shareholder value might *ex ante* (i.e. beforehand) justify cutting costs and accepting the risk of low-probability but very large environmental disasters (see Box 1.1 on The Deepwater Horizon oil spill). Even if such a disaster triggers legal actions that bankrupt the committing company, its shareholders are protected by limited liability and thus lose only the value of their shares. Box 1.5 illustrates the difficulties of enforcing legal standards, as companies control the information flows.

The key question remains how to rank shareholder and other stakeholder interests. Should shareholder interests come first, or should all interests be put on equal footing?

Box 1.5: The Dark Waters of DuPont

Company disasters might be discouraged by exposing directors to personal liability should they occur. But directors usually have liability insurance, which limits their personal exposure. Shapira and Zingales (2017) show how a respected company, like DuPont, willingly caused environmental damage by disposing a toxic chemical used in the making of Teflon in its West Virginia Plant. This case was turned into a legal thriller film called *Dark Waters*. The harmful pollution was a rational decision: under reasonable probabilities of detection, polluting was *ex ante* optimal from the company's perspective, albeit a very harmful decision from a societal perspective. Shapira and Zingales (2017) examine why different mechanisms of control, like legal liability, regulation and reputation, can all fail to deter socially harmful behaviour. One common reason for the failures of deterrence mechanisms is that the company controls most of the information and its release.

1.3.2 The Refined Shareholder Model

Although the shareholder model cannot fully satisfy the interests of stakeholders, there are also problems with the stakeholder model (Tirole, 2001). The manager has to serve all interests. Managers may, in that case, choose an objective function that is most closely relevant to their own interests (Jensen, 2002). Stakeholder theory may thus leave managers unaccountable, as optimising several objectives simultaneously is difficult to measure and control.

Jensen (2002) argues that shareholder value maximisation is best achieved in practice by catering to all stakeholders—an approach he calls Enlightened Value Maximisation. This view defends stakeholder interests as a means to the end goal of shareholder value maximisation. However, it fails to resolve the many situations of clear conflict between the interests of shareholders and different stakeholders. It also fails to value impacts that the company may inflict on more distant stakeholders, such as the environment.

Nonetheless, the approach has a single roughly measurable objective, refined shareholder value, while explicitly recognising that good relations with stakeholders can boost firm value by easing contracting costs and facilitating surplus creation. Companies put systems in place for energy and emissions management, sustainable purchasing, IT, building and infrastructure to enhance environmental standards and all kinds of diversity in employment. The underlying objective of these activities remains economic. Although introducing sustainability into business might generate positive side effects for some sustainability aspects, the main purpose is to reduce costs and business risks, to improve reputation and attractiveness for new or existing human talent, to respond to new customer demands and segments, and thereby to increase profits, market positions, competitiveness, and shareholder value in the short term. Business success is still evaluated from a purely economic point of view and remains focused on serving the business itself and its economic goals (Dyllick & Muff, 2016).

The formal objective function of the refined shareholder model is as follows:

$$\max V = FV + b \cdot SV + c \cdot EV \quad \text{with } 0 < b, c \ll 1 \quad (1.2)$$

where SV and EV represent the social and environmental value of the company, respectively. Because companies only consider negative social and environmental impacts to the extent relevant for maximising financial value, the refined shareholder model applies relatively low weights for social and environmental value, b and c .

Examples of such negative impacts are using child labour, unsafe work conditions, and/or pesticides in the production process. Innovations in technology (measurement, information technology, data management) and science (life cycle analyses, social life cycle analyses, environmentally extended input–output analysis, ecological economics) make the quantification and monetisation of social and environmental impacts possible (see Chap. 5 how these can be calculated). Box 1.6 provides an example of the use of pesticides in agriculture.

Box 1.6 Balancing the Negative Impact of Pesticides

An example of negative impact is the use of pesticides in agriculture. On the upside, pesticides combat unwanted insects and thus boost the unhindered growth of crops. This leads to higher profits for the farmer, thus improving FV . On the downside, pesticides may pollute the ground water, which now cannot be used for drinking water. This negative impact, measured as negative EV , is not reflected in the farmer's income, but can be quantified by applying the local drink water price to the quantity of affected ground water. Alternatively, when the ground water is still used for drinking water, the additional health care costs of the local people and the potential loss of income due to (chronical) illness can be calculated as SV . When the calculated negative impact of EV or SV becomes too large, the farmer should not use pesticides according to the refined shareholder model. The exact cut-off point depends on the weights b and c in Eq. (1.2), but these typically rise when public opinion turns negative on the use of pesticides.

1.3.3 The Stakeholder Model

The stakeholder model argues that managers should balance the interests of all stakeholders, which include financial agents (shareholders and debt holders) as well as social agents (consumers, workers, suppliers). Magill et al. (2015) show that a large firm typically faces endogenous risks that may have a significant impact on the workers it employs and the consumers it serves. These risks generate externalities (or impacts) on these stakeholders, which are not internalised by shareholders. As a result, in competitive equilibrium, there is underinvestment in the prevention of these risks.

Magill et al. (2015) suggest that this underinvestment problem can be alleviated if companies are instructed to maximise the total welfare of their stakeholders rather than shareholder value alone (shareholder equilibrium). The stakeholder equilibrium can be implemented by introducing new property rights (employee rights and consumer rights) and instructing managers to maximise the stakeholder value of the company (the value of these rights plus the shareholder value).

The formal objective function of the stakeholder model is as follows:

$$\max STV = FV + b \cdot SV \quad \text{with } b = 1 \quad (1.3)$$

where STV represents the stakeholder value of the company. The weight b indicates the importance of social value. In the standard case with $b = 1$, the financial and social value components are equally weighted. This indicates that companies have little scope for increasing value for one stakeholder at the expense of other stakeholders.

Tirole (2001) formulates three problems with serving various stakeholders in the stakeholder model. First, the stakeholder model may reduce pledgeable income (income available for financiers), as cash flows are distributed to various stakeholders. Second, it may lead to a less clear mission and fewer incentives for managers, as they have to serve multiple masters. Third, divided control among multiple stakeholders may lead to deadlock in decision-making.

The first two problems can be addressed by formulating an aggregate measure of value for the various stakeholders. Stakeholder value, defined in Eq. (1.3), provides such an aggregate measure. Another example is integrated value, defined in Eq. (1.4) below. Chapter 3 on corporate governance addresses the third problem of decision-making in a company serving multiple stakeholders.

1.3.4 The Integrated Model

While the stakeholder model incorporates social value alongside financial value into the company's objective, it does not deal with environmental value. Hart and Zingales (2017) argue that social and environmental impacts are not perfectly separable from production decisions. Therefore, companies face a choice in the degree of sustainability in their business model.

Schoenmaker and Schramade (2019) introduce integrated value, which combines financial, social, and environmental value in an integrated way. A *responsible company* maximises this integrated value in the interest of current and future stakeholders by managing and balancing profit (financial value) and impact (social and environmental value). The inclusion of future stakeholders who will face the consequences of (lack of) environmental actions today ensures that environmental impacts are incorporated. While the Hart-Zingales model argues that (prosocial) shareholders decide on corporate policy, the Schoenmaker-Schramade integrated model states that the managing board is accountable to all stakeholders. Chapter 3 addresses decision-making in a multiple stakeholder setting.

A new business language is emerging around 'the integrated value' of the company. Traditional financial reports record assets, liabilities, and profits solely on the basis of financial and manufactured capitals (financial value). Integrated financial reports broaden this range to six capitals by adding human and social capitals (social value), natural capital (environmental value), and intellectual capital (all three values); see Chap. 17 on integrated reporting. These capitals incorporate the social and environmental impacts and are expressed in money. This single language of integrated reporting enables managers to analyse the trade-offs for decision-making.

The concept of integrated value creation means that a company aims to optimise its financial, social, and environmental value in the long term. The optimisation requires a careful balancing of the three dimensions whereby interconnections and trade-offs are analysed, but none should deteriorate in favour of the others. The formal objective function for optimising the integrated value is:

$$\max IV = FV + b \cdot SV + c \cdot EV \quad \text{with } b, c = 1 \quad (1.4)$$

where IV represents the integrated value of the company. In the standard case with $b, c = 1$, the financial, social, and environmental value components are equally weighted. To make Eq. (1.4) operational, corporates need to calculate and balance the three values. Chapter 6 provides decision rules for this multicriteria decision problem. An important premise of these integrated decision rules is that companies cannot just improve FV to compensate for negative SV and EV .

Box 1.7 provides an example of a company that pursues integrated value creation. KPN, a Dutch telecom operator, communicates long-term goals with all stakeholders, including customers, employees, investors, society, and the environment and balances the goals. These goals are published in KPN's annual report. It should be noted that the production and publication of integrated balance sheets and integrated profit & loss accounts (IP&L) is work in progress (see Chap. 17). Standards for the new capitals are emerging, and auditing of the new information has just begun.

Box 1.7: Integrated Value Creation at KPN

Dutch telecom operator KPN could maximise short run return on invested capital (ROIC) by cutting operating costs (e.g. marketing costs for new customers) and capital investments (e.g. large investments in new network technology), which would look great for short-term-minded shareholders. However, it would also effectively kill its business, as ROIC would soon drop sharply as market share and product margins would fall. To restore market share, KPN would have to spend more than the initial costs and investments needed to pursue its long-term strategic goals.

The company therefore manages five goals: shareholders, customers, employees, society, and the environment. It has key performance indicators on all five and reports on each one of them, which should give a much better understanding of long-term value drivers than the old reporting system based on financial indicators did. In particular, the net promoter score (NPS) for customers is found to be very powerful. Figure 1.6 shows the importance of balancing the goals. This balanced approach puts KPN's business on a more solid and less volatile footing.

The integrated model and the sustainable development agenda are connected. By managing financial, social, and environmental resources in an integrated way, companies contribute to sustainable development at the global level. Figure 1.7 illustrates the key goals in both approaches.

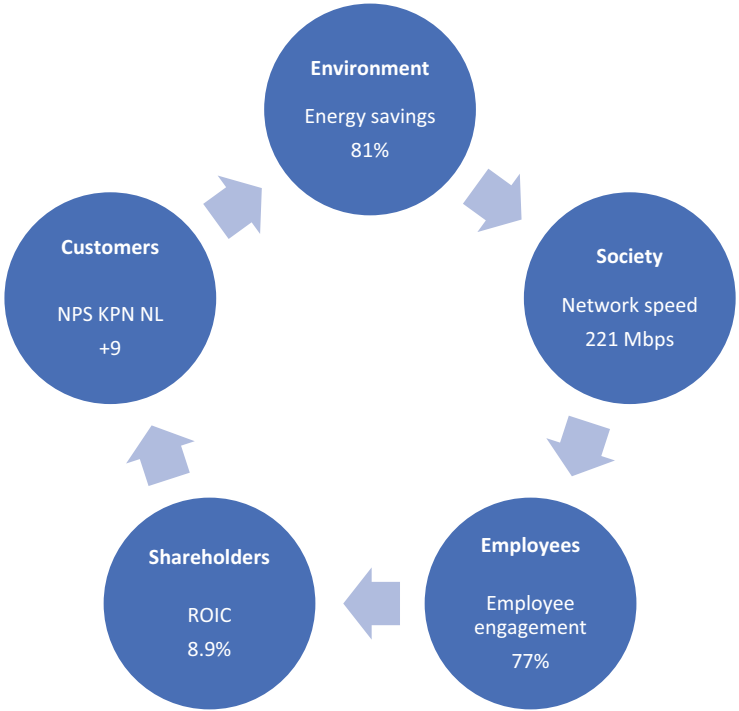


Fig. 1.6 Delivering value for current and future stakeholders. Note: Environment is measured as energy savings by customers; society as download speed of broadband fixed at mega-bits per second (Mbps); employees as employee engagement; shareholders as return on invested capital (ROIC); and customers as net promoter score (NPS). Source: KPN Integrated Annual Report 2018



Fig. 1.7 Goal of both the integrated model (company level) and sustainable development (global level)

1.4 Integration of Sustainability into Corporate Finance

An obvious way to deal with social and environmental issues is to put them in our economic models and methods. However, these models treat many of the social and environmental issues as *external impacts* or external effects (externalities), which affect other parties without these effects being reflected in market prices. Neo-classical models employ market prices as relevant signals for decision-making (e.g., investment, financing, production, or consumption decisions) and do not incorporate social and environmental impacts. Governments can use regulation or

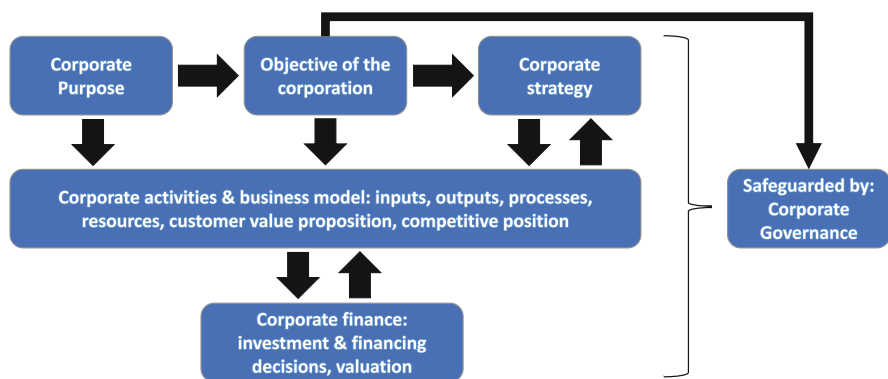


Fig. 1.8 Overview of the company

taxation to price or internalise external impacts. Societal forces can also put pressure on business to internalise social and environmental impacts (see Chap. 2).

Figure 1.8 provides an overview of the company that helps explain the set-up of this book *Corporate Finance for Long-Term Value*. At the top, in the middle, is the objective of the company. The integrated model argues that responsible companies serve the interests of current and future stakeholders. As a company moves from simply maximising financial value (FV) to optimising integrated value (IV), the question arises of what the company wants to achieve and where and how it can achieve most.

What are companies for? According to Friedman (1970), companies exist to maximise profitability, since that is what their owners (shareholders) want. In practice, however, companies are not just run or started for making as much money as possible. They operate in a societal context in which much more is expected from them than just making money. Often, companies are started because their founder identifies a large unmet need, or invented a brilliant technological innovation. For example, the founder of Lever Brothers (now Unilever) wanted to popularise cleanliness and hygiene in 1890s England. Companies exist for a purpose (Chap. 2; and Mayer, 2018). Based on that purpose, the company determines in whose interest it wants to achieve its objective: for shareholders, for current stakeholders, or for current and future stakeholders. Finally, the company sets its strategy and business model to meet societal needs with its products and/or services. Sustainability is an important driver of societal needs.

Disconnects between the owners or shareholders of a company, its managers, and the society in which it operates can and do happen. There is a key role for *corporate governance*, which refers to the mechanisms, relations, and processes by which a company is controlled and directed (Fig. 1.8 on the right; and Chap. 3). It involves balancing the many interests of the stakeholders of a company. Modern insights from corporate governance go beyond financial factors. As corporate ownership varies around the world, so do corporate governance challenges.

The remainder of the book addresses the investment and financing decisions that constitute the core of corporate finance. Its leading paradigm is integrated value, which combines financial, social, and environmental values. Applying this new paradigm is the real innovation of this corporate finance textbook. It uses the same basic methods and concepts, such as net present value, capital budgeting, valuation, and cost of capital, but adapts them to include social and environmental factors. The application of integrated value to these methods is illustrated with company examples throughout the book.

In sum, integrated value is relevant for companies in the following ways:

- Making investment decisions
- Measuring and reporting performance
- Conducting risk management
- Developing incentives
- Taking structural decisions (e.g. capital structure, M&As)

Part II of the book explains the basics of discount rates and investment decision rules. New decision rules based on integrated present value are introduced. Part III sets out the valuation of companies. Part IV addresses risk, return, and impact and derives the cost of integrated capital. Finally, Part V analyses corporate financial policies such as capital structure, payouts, reporting, and mergers and acquisitions in a multiple capital setting.

Applying Long-Term Value in Corporate Finance: Preview of the Book

At the end of this chapter, we show briefly what the expected effect of including social and environmental factors is on long-term value. This is basically a preview of the book.

The core model in corporate finance is the discounted cash flow (DCF) model to determine the value V of a project or a company:

$$V = \sum_{n=0}^N \frac{CF_n}{(1+r)^n}$$

whereby CF reflects the cash flows, r the discount rate (also called the cost of capital), and n the number of periods over which cash flows are discounted. The standard DCF model is used to calculate financial value FV .

Social (S) and environmental (E) issues can be added to the DCF model. As explained in Chap. 5, S and E issues can be expressed in their own units Q (e.g. life years saved by medical treatment or carbon emissions by using fossil fuels) and then multiplied by their respective shadow price SP derived from welfare theory. The shadow price for one life year, for example, is estimated at \$119,000 and the shadow

price per 1 ton of CO₂ equivalent is estimated at \$224 (IEF, 2022). The value flows *VF* are calculated as follows:

$$VF = Q \cdot SP$$

These value flows can be discounted with the DCF model to obtain *SV* and *EV*. It could be argued that cash flows are a special form of value flows expressed in cash. Here, we use the more general term of value flows to calculate integrated value *IV*:

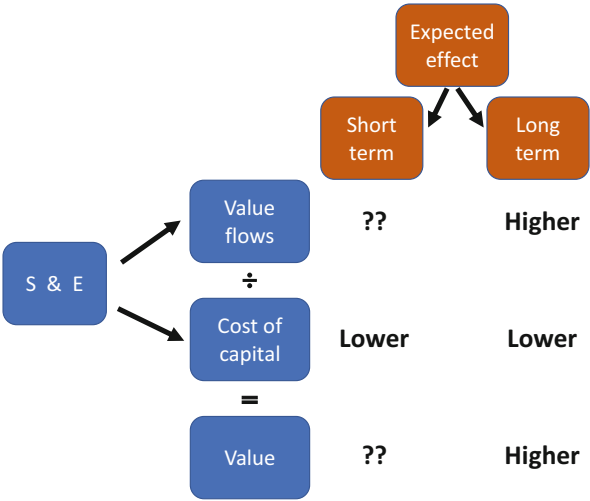
$$IV = \sum_{n=0}^N \frac{VF_n}{(1+r)^n}$$

Value Flows

Sustainability transitions result in changes in the markets for products and services. In the process of transition, social and environmental externalities are internalised. Internalisation means that the burdens of externalities are increasingly shifted back from society to the companies (and consumers) who cause them. Companies that create FV at the expense of SV or EV (social and environmental externalities) will often be affected with lower FV if and when internalisation occurs. By contrast, companies that provide solutions for solving negative SV and EV are rewarded with stronger FV (see, for example, Kurznack et al. (2021) in Chap. 2).

The alignment between profit (FV) and impact (SV and EV) depends on one key assumption, namely that sustainability transitions will happen at some point in time. The timing of transition—early or late—is difficult to predict. The investments for transitions are done today or in the near future, while the timing of the benefits is uncertain. The expected effect of sustainability improvements (S and E) on value flows is thus uncertain in the short term, but likely to be positive in the long term (see Fig. 1.9).

Fig. 1.9 Expected effect of improving S and E on value



Cost of Capital

The cost of capital increases with negative social and environmental externalities (because of a risk premium for the systemwide dimension of social and environmental risk) and decreases with positive social and environmental impact (because of reduced risk). Negative externalities effectively increase the leverage of integrated capital. Chapter 12 provides the emerging evidence for the relevance of E and S to the cost of capital. In a global study covering 77 countries, Bolton and Kacperczyk (2023), for example, found a positive and significant relationship for short-term and long-term measures of carbon transition risk and return: higher risk leads to a higher cost of capital because of a positive risk premium. In a similar way, Hong and Kacperczyk (2009) found a positive risk premium for sin stocks, such as alcohol, tobacco, and gaming.

We can now turn the argument around. Lower S and E risk, which means better S and E performance, leads to a lower cost of capital in the short term and in the long term (see Fig. 1.9).

Value Effect

For positive S and/or E impacts, higher value flows (in the numerator) and a lower cost of capital (in the denominator) are expected to produce higher company value in Fig. 1.9. And vice versa for negative impacts.

So, companies with a positive impact are likely to produce more long-term value. In the long run, financial, social, and environmental value are largely aligned. The challenge lies in trade-offs across time and between types of value, which can interact in numerous ways. In the remainder of this book, we show how the nexus between impact and long-term value works for the various corporate finance methods.

1.5 Conclusions

Coming from an ‘empty’ world with abundant natural resources, the Industrial Revolution brought prosperity in the form of economic and population growth. At the same time, this growth created social and environmental challenges. To address these issues, the United Nations developed the Sustainable Development Goals for 2030. Sustainability means that current and future generations have the resources needed, such as food, water, healthcare, and energy, without stressing the Earth system processes.

Corporate Finance for Long-Term Value looks at how corporate finance (investment decisions, valuation, reporting) interacts with economic, social, and environmental issues. This chapter shows how corporate finance has the potential to move from finance as a goal (shareholder value) to finance as a means towards integrated value creation. In his book *Finance and the Good Society*, Shiller (2012) provides some stimulating examples of how finance can serve society and its citizens.

We are in transition to a low-carbon and more circular economy. The impacts of the current carbon-intensive economy are becoming increasingly clear to the wider public. Examples are more catastrophic weather events, droughts, flooding in countries close to the equator, and air pollution. A case in point is California, where air pollution from heavy traffic in the 1990s prompted environmental regulations and stimulated innovations, such as the electric cars of Tesla and solar technology. Finance is about anticipating such events and incorporating expectations into today's valuations for investment decisions. Finance can thus contribute to a swift transition to a low-carbon economy.

Key Concepts Used in This Chapter

Corporate governance refers to the mechanisms, relations, and processes by which a company is controlled and directed. It involves balancing the many interests of the stakeholders of a company.

Environmental issues or ecological issues are abiotic or biotic issues that influence living organisms; see *planetary boundaries* for the most critical ecological issues

Impacts (also called external impacts or externalities) refer to consequences of activities that affect other (or third) parties without this being reflected in market prices

Integrated model means that a company should balance or optimise the interests of its current and future stakeholders: customers, employees, suppliers, shareholders, the community, and the environment

Integrated value is obtained by combining the financial, social, and environmental value in an integrated way (with regard for the interconnections)

Linear production and consumption system is based on the extraction of raw materials (take), processing into products (make), consumption (use), and disposal (waste)

Living wage is a wage for a full-time worker sufficient to provide their family's basic needs for an acceptable standard of living; a living wage varies with the local cost of living

Planetary boundaries framework consists of nine planetary boundaries within which humanity can continue to develop and thrive for generations to come; these boundaries include climate change, biosphere integrity, land-system change, freshwater change, biochemical flows, ocean acidification, atmospheric aerosol loading, stratospheric ozone depletion, and novel entities (plastics and chemicals)

Precautionary principle states that an action should not be taken (or a boundary should not be crossed) if the consequences are uncertain and potentially dangerous

Resilience of a system (e.g. an ecosystem or organisation) is the adaptive capacity of a system to deal with unpredictable shocks

Responsible company manages and balances profit (financial value) and impact (social and environmental value)

Resource abundance refers to the plentiful availability of natural resources such as minerals, metal ores, fossil fuels, land, and freshwater

Shareholder model means that the ultimate measure of a company's success is the extent to which it enriches its shareholders

Social foundations consist of the twelve top social priorities, grouped into three clusters, focused on enabling people to be (1) well: through food security, adequate income, improved water and sanitation, housing and healthcare; (2) productive: through education, decent work and modern energy services; and (3) empowered: through networks, gender equality, social equity, having political voice and peace and justice

Stakeholder model means that a company should balance or optimise the interests of all its stakeholders: customers, employees, suppliers, shareholders, and the community

Sustainability means that current and future generations have the resources needed, such as food, water, healthcare and energy, without stressing processes within the Earth system

Suggested Reading

- Common, M., & Stagl, S. (2012). *Ecological economics: An introduction*. Cambridge University Press.
- Dyllick, T., & Muff, K. (2016). Clarifying the meaning of sustainable business introducing a typology from business-as-usual to true business sustainability. *Organization and Environment*, 29(2), 156–174.
- Hart, O., & Zingales, L. (2017). Companies should maximize shareholder welfare not market value. *Journal of Law, Finance, and Accounting*, 2(2), 247–274.
- Sachs, J. (2015). *The age of sustainable development*. Columbia University Press.
- Steffen, W., Richardson, K., Rockström, J., Cornell, S. E., Fetzer, I., Bennett, E. M., Biggs, R., Carpenter, S. R., de Vries, W., de Wit, C. A., Folke, C., Gerten, D., Heinke, J., Mace, G. M., Persson, L. M., Ramanathan, V., Reyers, B., & Sörlin, S. (2015). Planetary boundaries: Guiding human development on a changing planet. *Science*, 347(6223), 736–747.
- United Nations. (2015). UN Sustainable Development Goals (UN SDGs)—Transforming our world: the 2030 agenda for sustainable development, A/RES/70/1, New York.

References

- Acemoglu, D., & Robinson, J. (2012). *Why nations fail*. Crown Business.
- Barone, G., & Mocetti, S. (2016). Inequality and trust: new evidence from panel data. *Economic Inquiry*, 54(2), 794–806.
- Berger, P. (2018). *The limits of social cohesion: Conflict and mediation in pluralist societies*. Routledge.
- Berk, J., & DeMarzo, P. (2017). *Corporate finance: The core* (4th ed.). Pearson.
- Bolton, P., & Kacperczyk, M. (2023). Global pricing of carbon-transition risk. *Journal of Finance*, forthcoming.
- Brealey, R., Myers, S., & Allen, F. (2020). *Principles of corporate finance* (13th ed.). McGraw-Hill.
- Brundtland Report. (1987). *Our common future*. The United Nations World Commission on Environment and Development, United Nations.
- Cambridge Institute for Sustainability Leadership (CISL). (2021). *Handbook for nature-related financial risks: key concepts and a framework for identification*.

- Daly, H., & Farley, J. (2011). *Ecological economics: Principles and applications*. Island Press.
- Dyllick, T., & Muff, K. (2016). Clarifying the meaning of sustainable business introducing a typology from business-as-usual to true business sustainability. *Organization and Environment*, 29(2), 156–174.
- Friedman, M. (1970). The social responsibility of business is to increase its profits. *The New York Times Magazine*, 13 September.
- Hart, O., & Zingales, L. (2017). Companies should maximize shareholder welfare not market value. *Journal of Law, Finance, and Accounting*, 2(2), 247–274.
- Holling, C. S. (2001). Understanding the complexity of economic, ecological, and social systems. *Ecosystems*, 4(5), 390–405.
- Hong, H., & Kacperczyk, M. (2009). The price of sin: The effects of social norms on markets. *Journal of Financial Economics*, 93(1), 15–36.
- Impact Economy Foundation (IEF). (2022). *Impact-weighted accounts framework*.
- Intergovernmental Panel on Climate Change. (2023). *Synthesis report of the IPCC sixth assessment report (AR6)*.
- Jensen, M. (2002). Value maximization, stakeholder theory, and the corporate objective function. *Business Ethics Quarterly*, 12(2), 235–256.
- Kurzack, L., Schoenmaker, D., & Schramade, W. (2021). A model of long-term value creation. *Journal of Sustainable Finance & Investment*, 1–19.
- Magill, M., Quinzii, M., & Rochet, J.-C. (2015). A theory of the stakeholder corporation. *Econometrica*, 83(5), 1685–1725.
- Mayer, C. (2018). *Prosperity: Better business makes the greater good*. Oxford University Press.
- Meadows, D. H., Meadows, D. L., Randers, J., & Behrens, W. W., III. (1972). *Limits to economic growth: A report for the Club of Rome's project on the predicament of mankind*. Universe Books.
- Mercer. (2015). *Investing in a time of climate change*.
- National Commission. (2011). *Deep water: The Gulf Oil disaster and the future of offshore drilling*. National Commission on the BP Deepwater Horizon Oil Spill and Offshore Drilling, US Government.
- Norström, A. V., Dannenberg, A., McCarney, G., Milkoreit, M., Diekert, F., Engström, G., Fishman, R., Gars, J., Kyriakopoulou, E., Manoussi, V., Meng, K., Metian, M., Sanctuary, M., Schlüter, M., Schoon, M., Schultz, L., & Sjöstedt, M. (2014). Three necessary conditions for establishing effective sustainable development goals in the anthropocene. *Ecology and Society*, 19(3), 8.
- Persson, L., Carney Almroth, B., Collins, C., Cornell, S., de Wit, C. A., Diamond, M., Fantke, P., Hassellöv, M., MacLeod, M., Ryberg, M., Sogaard Jørgensen, P., Villarrubia-Gómez, P., Wang, Z., & Hauschild, M. (2022). Outside the safe operating space of the planetary boundary for novel entities. *Environmental Science & Technology*, 56(3), 1510–1521.
- Raworth, K. (2017). *Doughnut economics: Seven ways to think like a 21st-century economist*. Random House Business Books.
- Robinson, A., Calov, R., & Ganopolski, A. (2012). Multistability and critical thresholds of the Greenland ice sheet. *Nature Climate Change*, 2(6), 429–432.
- Rockström, J., & Sukhdev, P. (2016). *How food connects all the SDGs*. Stockholm Resilience Centre.
- Sachs, J. (2015). *The age of sustainable development*. Columbia University Press.
- Schoenmaker, D., & Schramade, W. (2019). *Principles of sustainable finance*. Oxford University Press.
- Shapira, R., & Zingales, L. (2017). *Is pollution value-maximizing? The DuPont Case* (CEPR Discussion Paper, DP12323).
- Shiller, R. (2012). *Finance and the good society*. Princeton University Press.
- Steffen, W., Richardson, K., Rockström, J., Cornell, S., Fetzer, I., Bennett, E., Biggs, R., Carpenter, S., de Vries, W., de Wit, C., Folke, C., Gerten, D., Heinke, J., Mace, G., Persson, L.,

- Ramanathan, V., Reyers, B., & Sörlinet, S. (2015). Planetary boundaries: Guiding human development on a changing planet. *Science*, 347(6223), 736–747.
- Tirole, J. (2001). Corporate governance. *Econometrica*, 69(1), 1–35.
- United Nations. (2015). UN Sustainable Development Goals (UN SDGs)—Transforming our world: the 2030 Agenda for Sustainable Development, A/RES/70/1, New York.
- United Nations Framework Convention on Climate Change. (2015). Adoption of the Paris Agreement.
- Wang-Erlandsson, L., Tobian, A., van der Ent, R. J., Fetzer, I., te Wierik, S., Porkka, M., Staal, A., Jaramillo, F., Dahlmann, H., Singh, C., Greve, P., Gerten, D., Keys, P. W., Gleeson, T., Cornell, S. E., Steffen, W., Bai, X., & Rockström, J. (2022). A planetary boundary for green water. *Nature Reviews Earth & Environment*, 3, 382–390.

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